

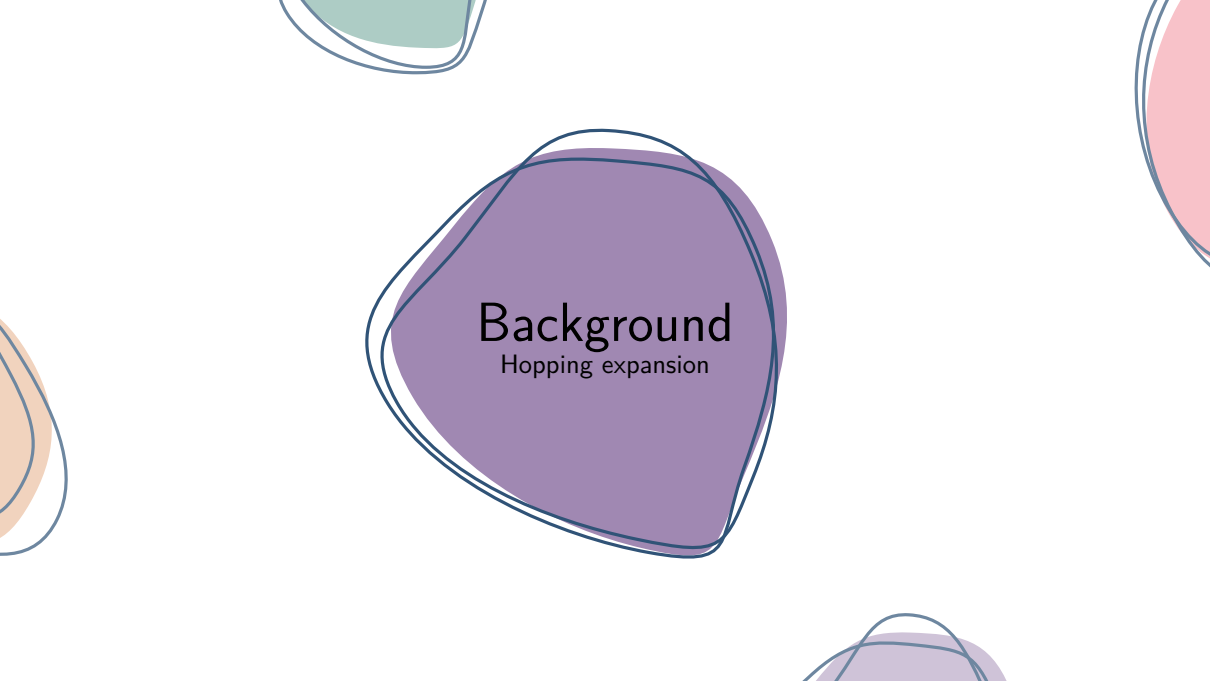
# Neural-network approaches for preconditioning the Dirac equation in lattice QCD

**Simon Pfahler**

University of Regensburg

based on arXiv:2602.23840 [hep-lat]

May 26, 2026

The background features several abstract, hand-drawn shapes in muted colors: a teal shape at the top left, a pink shape at the top right, an orange shape on the left edge, and a purple shape at the bottom right. The central focus is a large, irregular purple shape with a dark blue outline, which contains the text.

# Background

Hopping expansion

# Problem statement

## Bigger picture

- ▶ Lattice QCD calculations are expensive
- ▶ One of the most expensive parts is solving the Dirac equation
- ▶ Preconditioners can accelerate this

## Preconditioners

- ▶ State of the art: algebraic multigrid (AMG)
- ▶ But AMG needs costly setup
- ▶ Can this setup be avoided?
- ▶ **Goal:** Preconditioner competitive with AMG, but with very little (or even no) setup

# Lattice Definitions 1

- ▶ We simulate QCD on a discrete space-time lattice  
→ fermion fields  $\psi \in \mathbb{C}^{V \times 4 \times 3}$   
→ gauge fields  $U \in SU(3)^{4 \times V}$
- ▶ Gauge transformations  $\Omega \in SU(3)^V$  transform the fields as

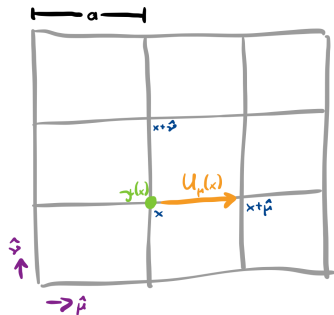
$$\psi(x) \rightarrow \Omega(x)\psi(x)$$

$$U_\mu(x) \rightarrow \Omega(x)U_\mu(x)\Omega^\dagger(x + \hat{\mu})$$

- ▶ This leads to parallel-transport operators

$$H_\mu\psi(x) = U_\mu^\dagger(x - \hat{\mu})\psi(x - \hat{\mu})$$

These transform as  $H_\mu\psi(x) \rightarrow \Omega(x)H_\mu\psi(x)$



## Lattice Definitions 2

Dirac equation:  $D\psi = \phi$

We use the Wilson(-clover) Dirac operator

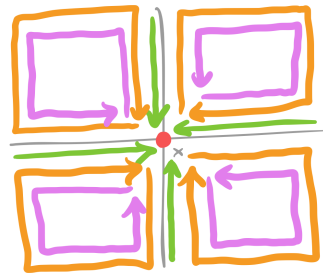
$$D_{WC} = D_W - \frac{c_{sw}}{4} \sum_{\mu, \nu} \sigma_{\mu\nu} F_{\mu\nu}$$

$$D_W = (m + 4) - \sum_{\mu} \left( \frac{1 - \gamma_{\mu}}{2} H_{-\mu} + \frac{1 + \gamma_{\mu}}{2} H_{+\mu} \right)$$

$$F_{\mu\nu} = \frac{1}{8} (Q_{\mu\nu} - Q_{\nu\mu})$$

$$Q_{\mu\nu} = P_{+\nu, +\mu} + P_{-\mu, +\nu} + P_{-\nu, -\mu} + P_{+\mu, -\nu}$$

$$P_{\pm\mu, \pm\nu} = H_{\mp\nu} H_{\mp\mu} H_{\pm\nu} H_{\pm\mu}$$



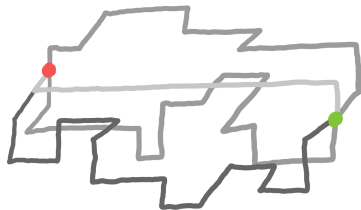
action of  $D_{WC}$  on  $\psi(x)$

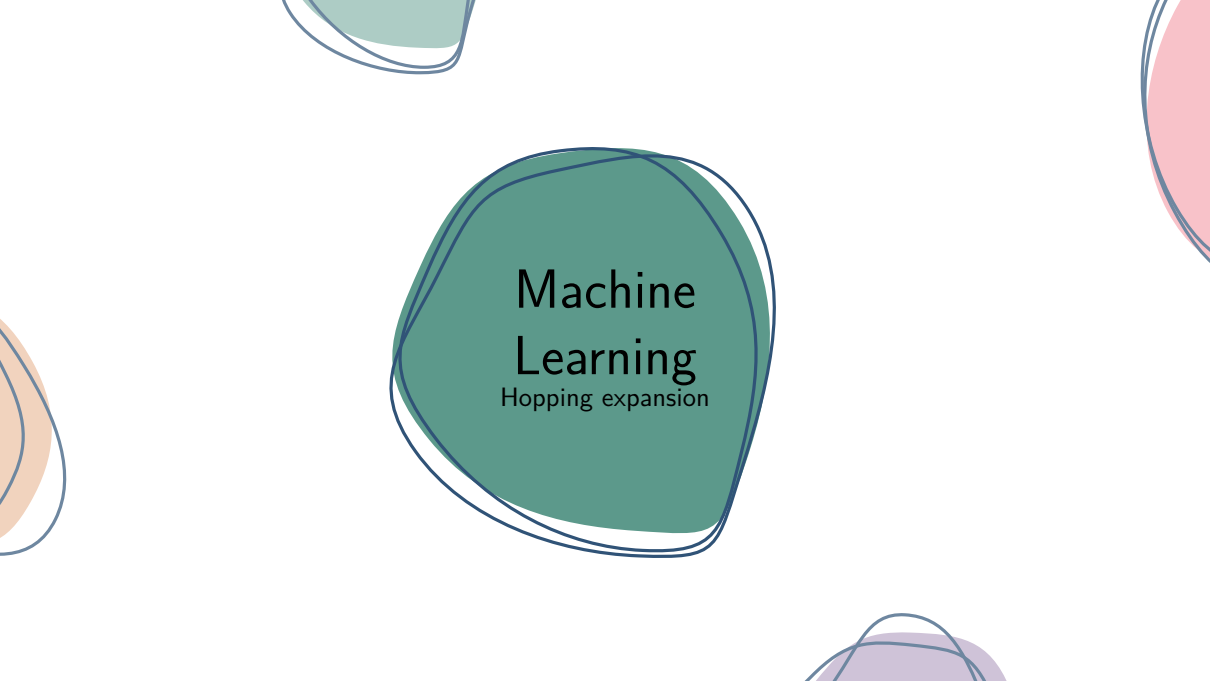
# Hopping expansion

- ▶ We can write  $D_W = (m + 4)(\mathbb{1} - \kappa H)$  with hopping parameter  $\kappa = \frac{1}{2(m+4)}$
- ▶ For  $\kappa < \frac{1}{8}$ , the inverse is of  $D_W$  is given by

$$D_W^{-1} = \frac{1}{m+4}(\mathbb{1} - \kappa H)^{-1} = \frac{1}{m+4} \sum_{n=0}^{\infty} \kappa^n H^n$$

- ▶ Can we learn systematic hopping-style expansions using Machine Learning?  
 $\Rightarrow$  Yes, let's see how!



The background is white with several large, hand-drawn, overlapping circles in muted colors: teal, orange, pink, and purple. The central teal circle contains the text.

# Machine Learning

Hopping expansion

# Network Architecture



## Dense linear (L) layer

- ▶  $n$  spin-color fields  
 $\varphi \in \mathbb{C}^{L_x \times L_y \times L_z \times L_t \times N_c \times N_s}$
- ▶ build  $m$  linear combinations of the input fields, with spin matrices  $W_{ij} \in \mathbb{C}^{N_s \times N_s}$  as weights:

$$\psi_i(x) \stackrel{\text{L}}{=} \sum_{j=1}^n W_{ij} \varphi_j(x)$$

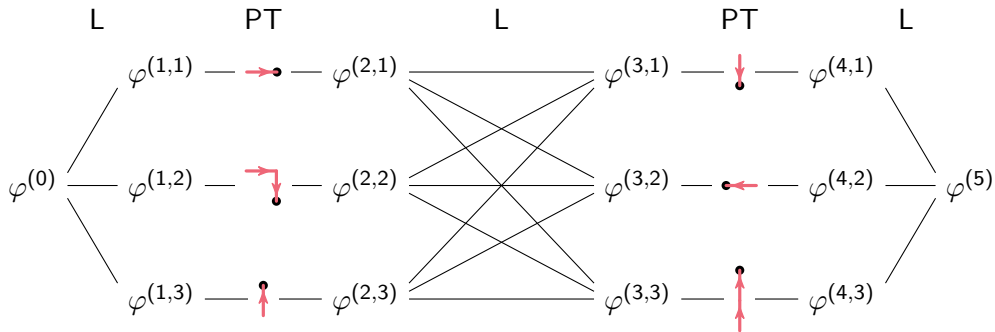
## Parallel Transport (PT) layer

- ▶  $n$  spin-color fields  
 $\varphi \in \mathbb{C}^{L_x \times L_y \times L_z \times L_t \times N_c \times N_s}$
- ▶ apply a different parallel transport  $p^{(i)}$  to every input field:

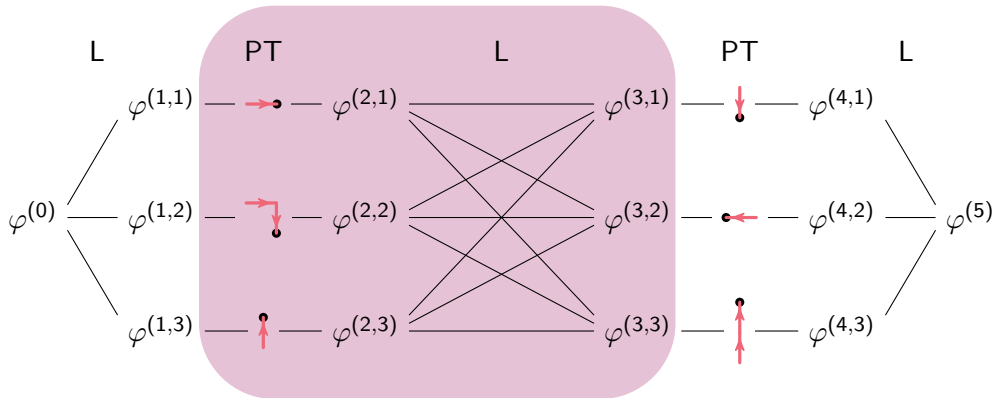
$$\psi_i(x) \stackrel{\text{PT}}{=} T_{p^{(i)}} \varphi_i(x)$$



# Network Architecture



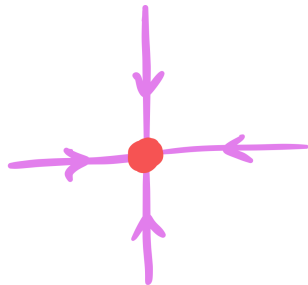
# Network Architecture



## Choice of PT paths

The most natural choice of PT paths is  
only nearest-neighbor hops:

$$P = \{1\} \cup \{H_p | p \in \{\pm 1, \pm 2, \pm 3, \pm 4\}\}$$



# Score function



- ▶ To train our network, we need a score function
- ▶ Ideally, we want to optimize the condition number of  $MD$ ,

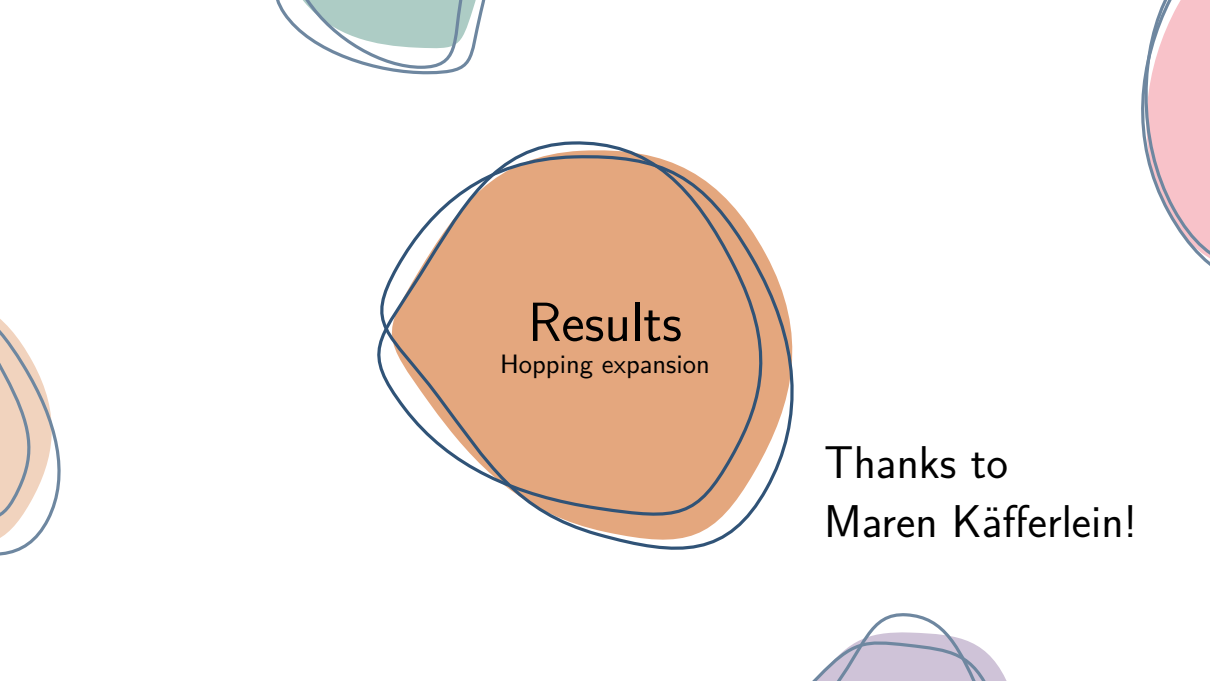
$$\kappa(MD) = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

→ Expensive and unstable ☹

- ▶ Instead, we use the surrogate cost function

$$C = \|MD\varphi - \varphi\|^2$$

with random fermion fields  $\varphi$

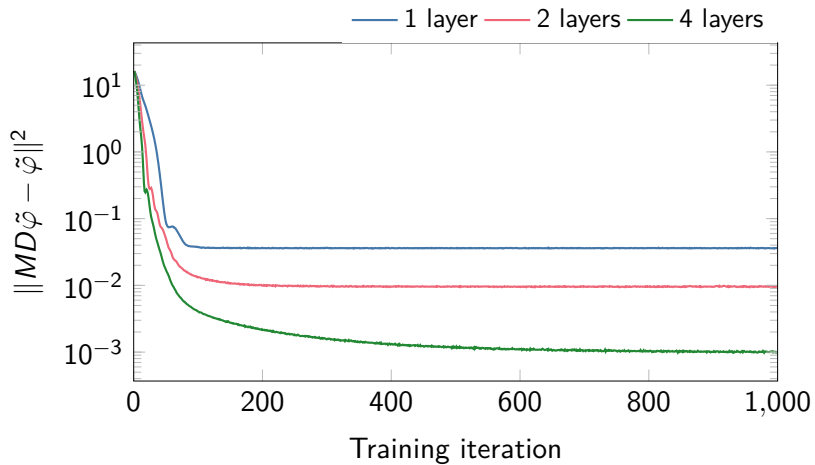
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# Results

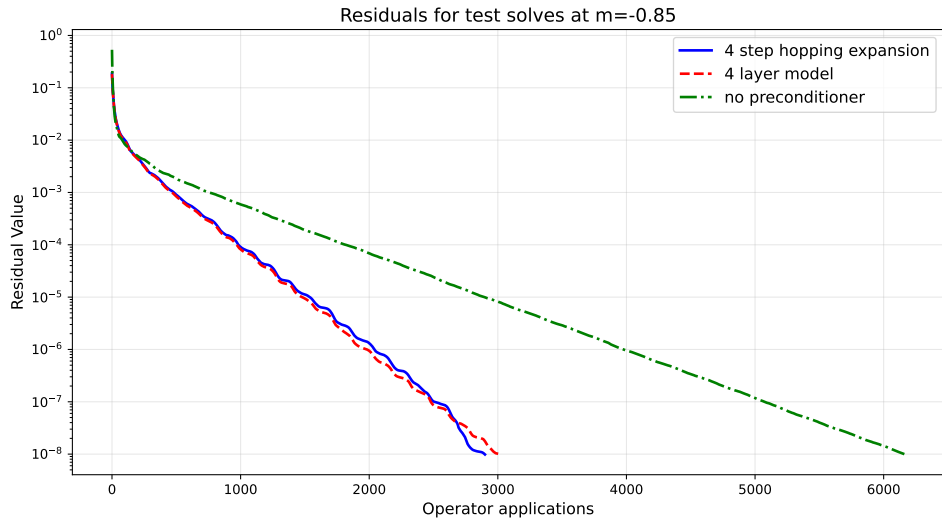
Hopping expansion

Thanks to  
Maren Käfferlein!

# Training



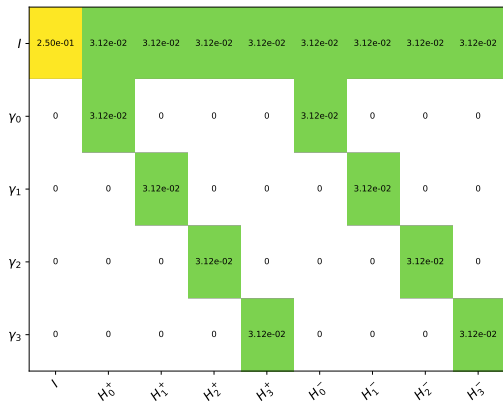
# Test solves



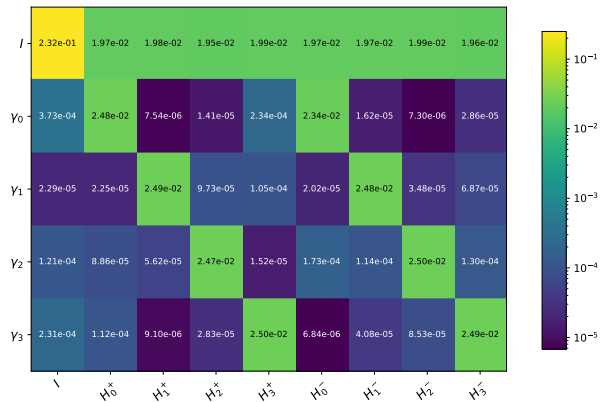
# Analytical structure

1 layer

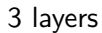
Hopping expansion

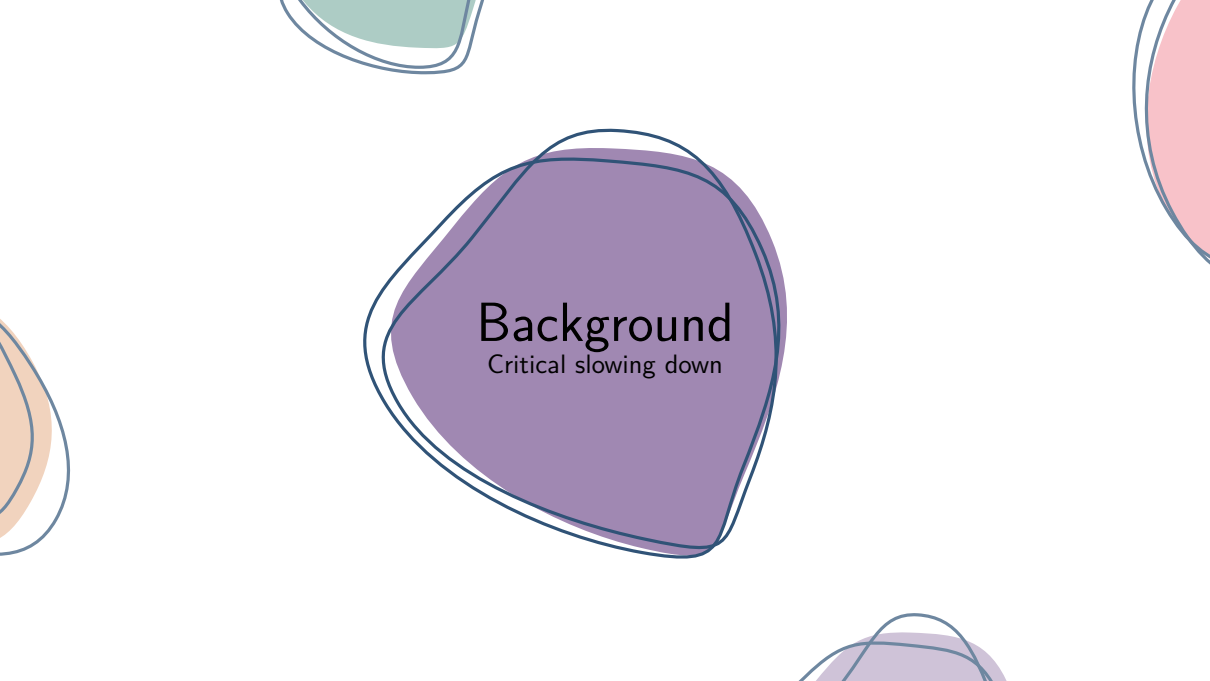


Learned coefficients







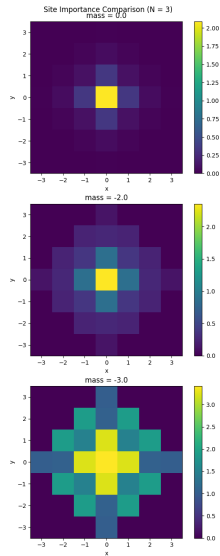
The background features several large, overlapping, organic shapes in muted colors: teal, orange, pink, and purple. Each shape is outlined with a thin, dark blue line. The central purple shape is the largest and contains the text.

# Background

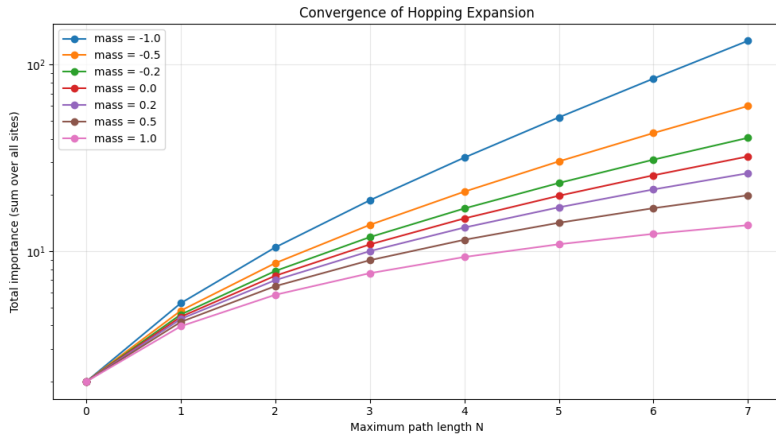
Critical slowing down

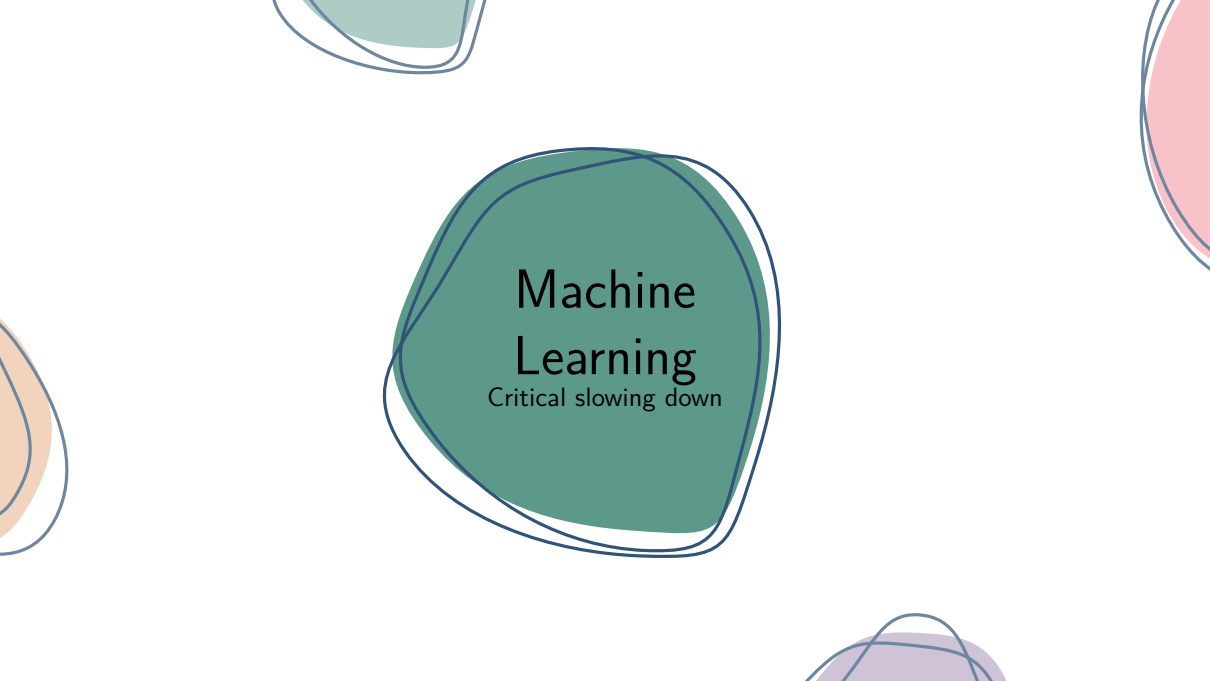
# Critical slowing down

- ▶ For  $\kappa = \frac{1}{2(m+4)} < \frac{1}{8}$ , the hopping expansion does not converge anymore, as long paths dominate
- ▶ Near  $\kappa \approx \frac{1}{8}$ , we need an increasing amount of layers to build a good approximation of  $D^{-1}$
- ▶ How can we make our networks more efficient in this regime?



# Critical slowing down

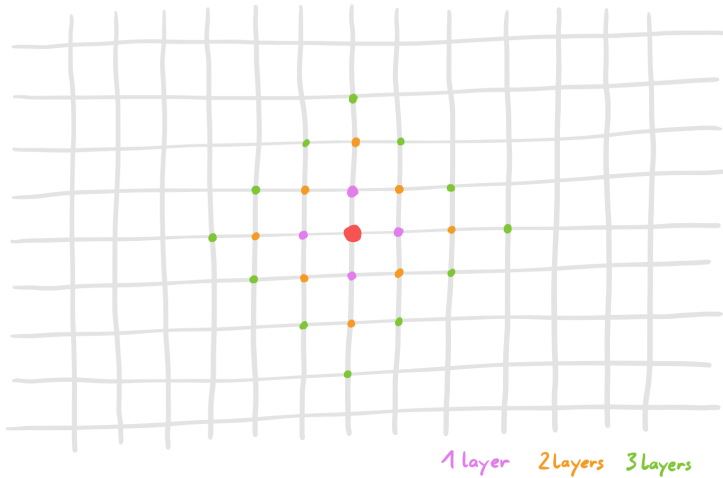


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# Machine Learning

Critical slowing down

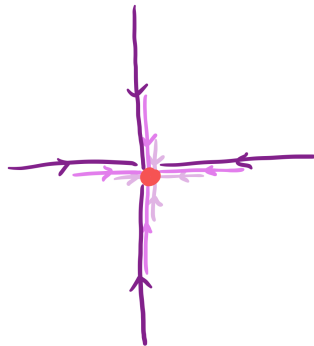
# PT Paths 1



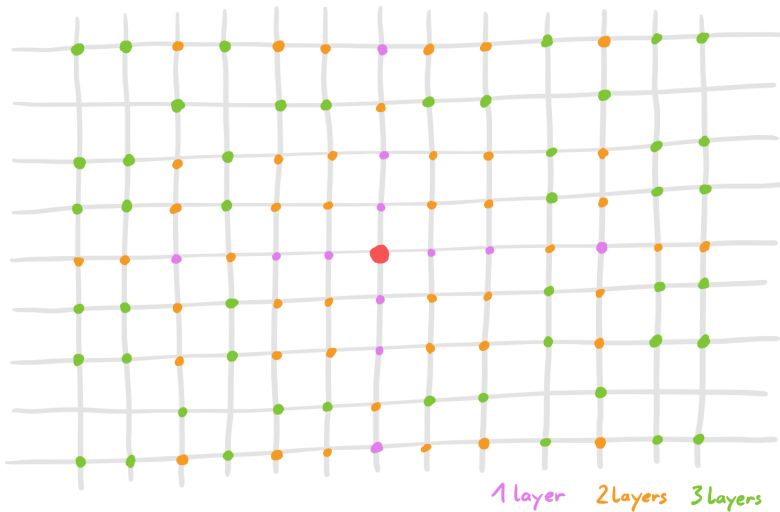
## PT Paths 2

To allow for longer paths without increasing the number of paths too much, we choose

$$P = \{1\} \cup \{H_p^n | p \in \{\pm 1, \pm 2, \pm 3, \pm 4\}, \\ n \in \{2^0, 2^2, \dots, L_p/2\}\}$$



# PT Paths 3





# Score function

## ► Standard choice:

$$C = \|MD\varphi - \varphi\|^2$$

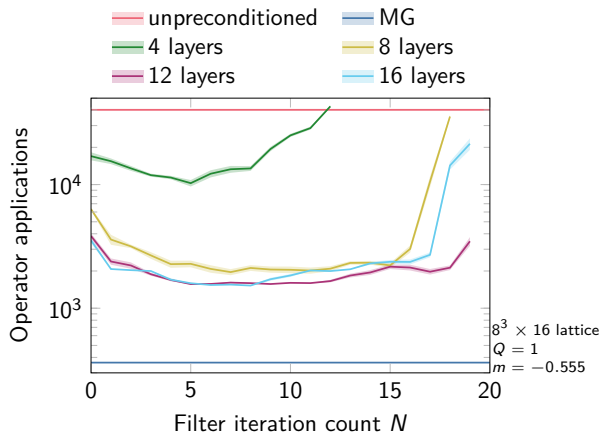
with random field  $\varphi$

## ► Filtered cost function:

$$C = \|MD\tilde{\varphi} - \tilde{\varphi}\|^2$$

where  $\tilde{\varphi}$  is obtained through  $N$  iterations of GMRES for  $D\varphi = 0$  with random initial guess

⇒ Stronger emphasis on the low modes!



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# Results

Critical slowing down

# Gauge configurations

$8^3 \times 16$  ensemble<sup>1</sup>:

- ▶ Wilson action
- ▶ quenched, heat bath
- ▶ Wilson-clover Dirac operator
- ▶ configurations with topological charge  $|Q| \in \{0, 1, 2\}$

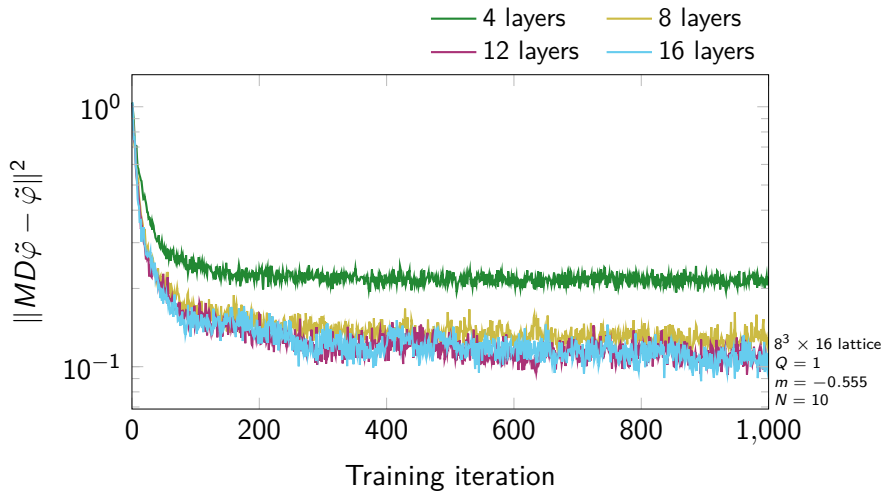
$16^3 \times 32$  ensemble<sup>2</sup>:

- ▶ Wilson action
- ▶ quenched, HMC
- ▶ Wilson-clover Dirac operator
- ▶ configurations with topological charge  $|Q| \in \{0, 4\}$

<sup>1</sup> [doi.org/10.5281/zenodo.15018324](https://doi.org/10.5281/zenodo.15018324)

<sup>2</sup> [doi.org/10.5281/zenodo.17494829](https://doi.org/10.5281/zenodo.17494829)

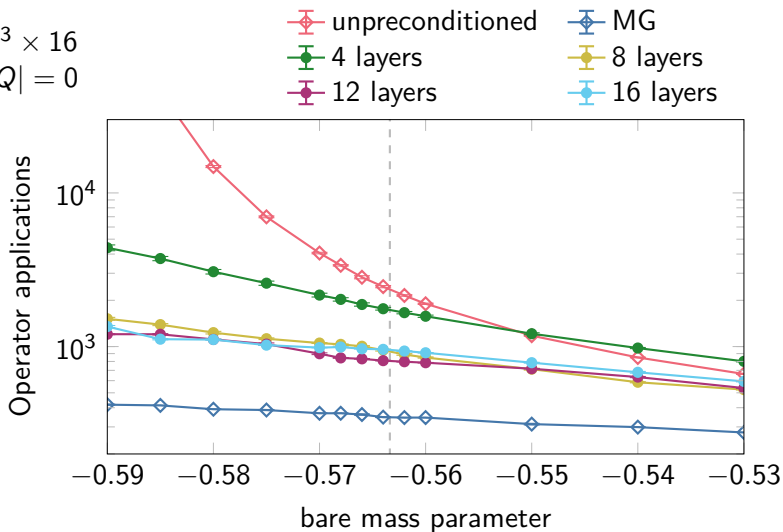
# Training cost



# Critical slowing down

$$8^3 \times 16$$

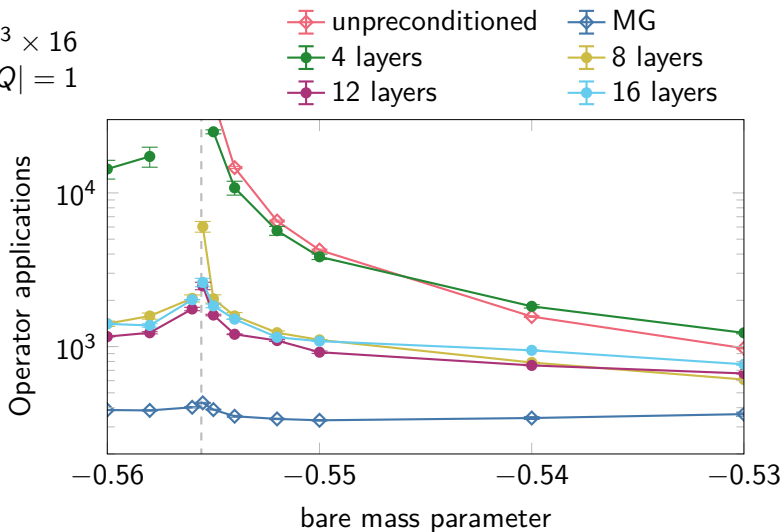
$$|Q| = 0$$



# Critical slowing down

$$8^3 \times 16$$

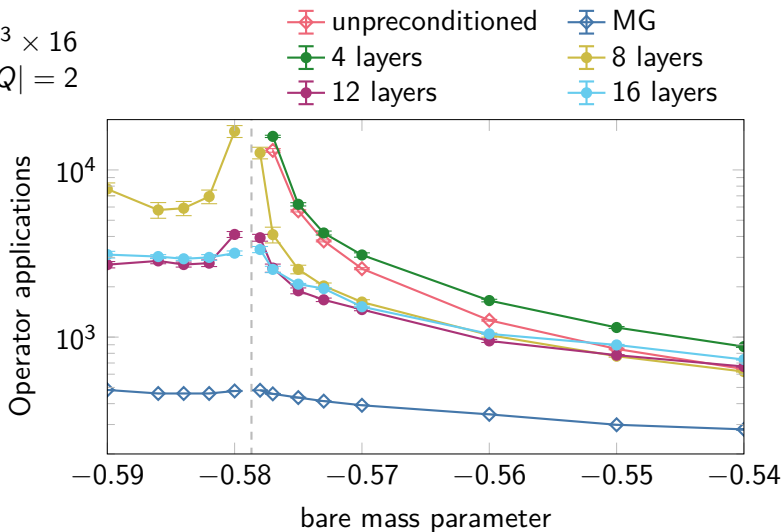
$$|Q| = 1$$



# Critical slowing down

$$8^3 \times 16$$

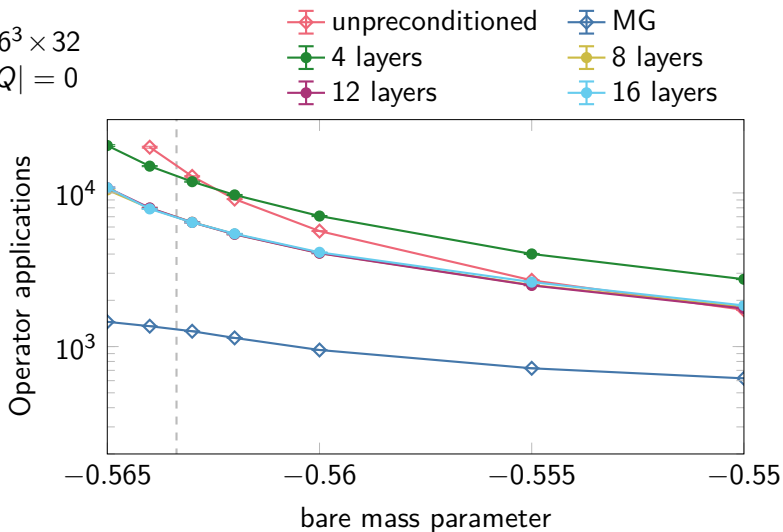
$$|Q| = 2$$



# Critical slowing down

$16^3 \times 32$

$|Q| = 0$

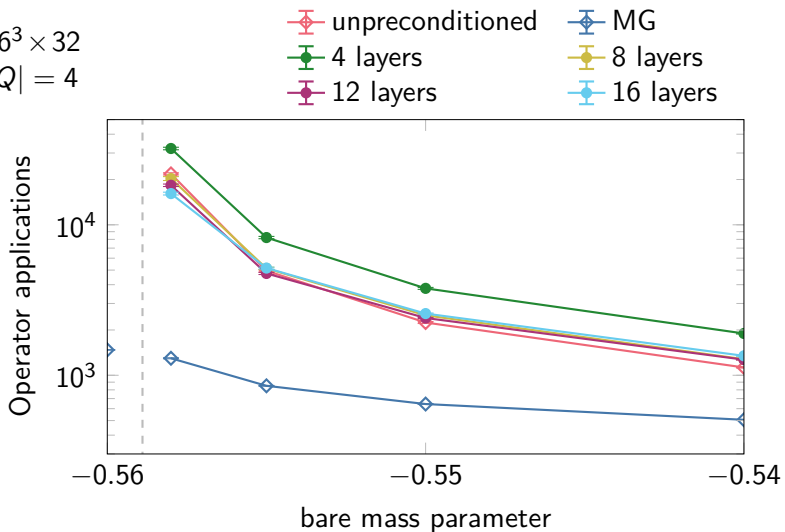




# Critical slowing down

$16^3 \times 32$

$|Q| = 4$



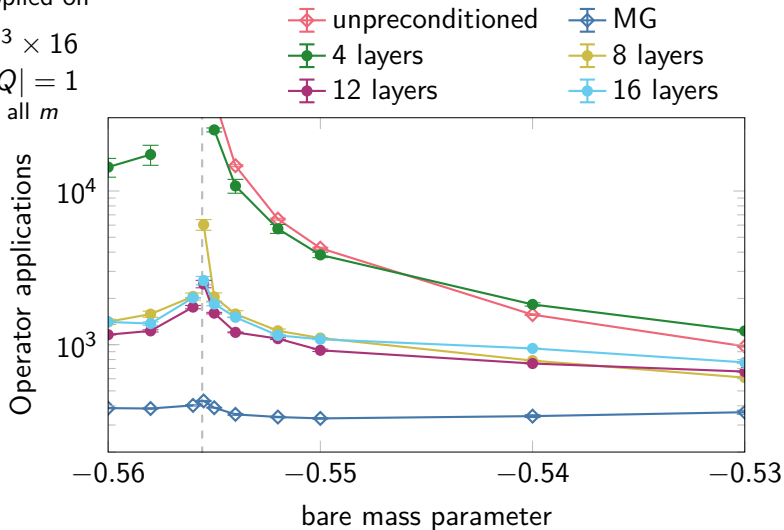
# Transfer

trained and  
applied on

$$8^3 \times 16$$

$$|Q| = 1$$

all  $m$



# Transfer

trained on

$$8^3 \times 16$$

$$|Q| = 0$$

$$m = -0.56$$

applied on

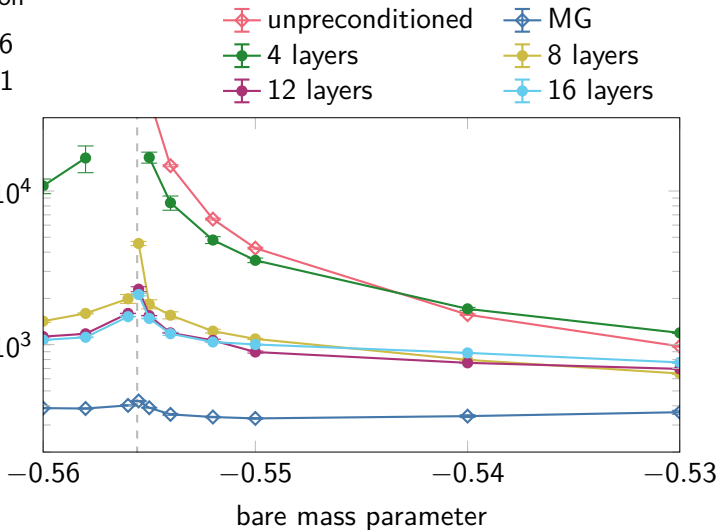
$$8^3 \times 16$$

$$|Q| = 1$$

$$\text{all } m$$

$\rightarrow$

Operator applications



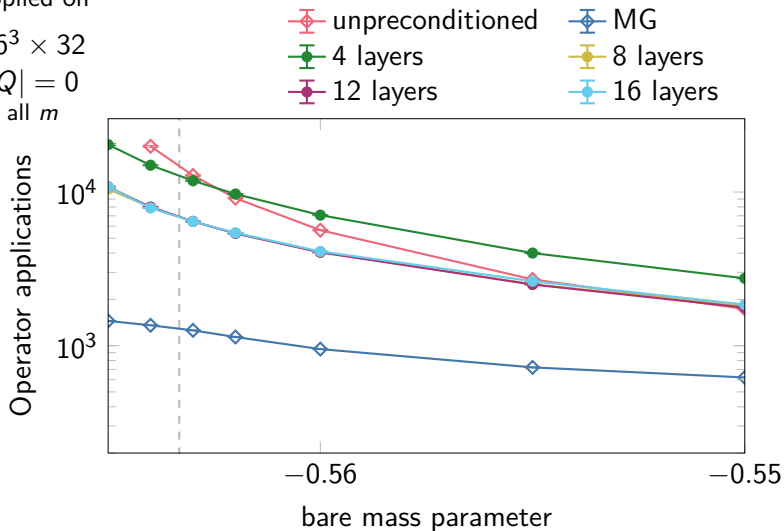
# Transfer

trained and  
applied on

$$16^3 \times 32$$

$$|Q| = 0$$

all  $m$



# Transfer

trained on

$$8^3 \times 16$$

$$|Q| = 0$$

$$m = -0.56$$

applied on

$$16^3 \times 32$$

$$|Q| = 0$$

$$\text{all } m$$

→

unpreconditioned

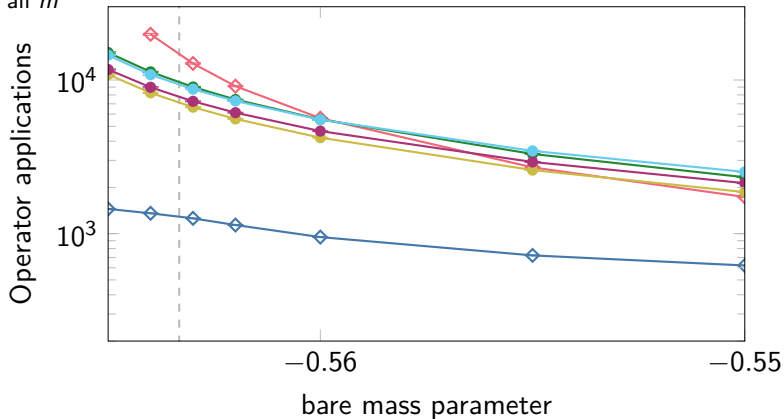
4 layers

12 layers

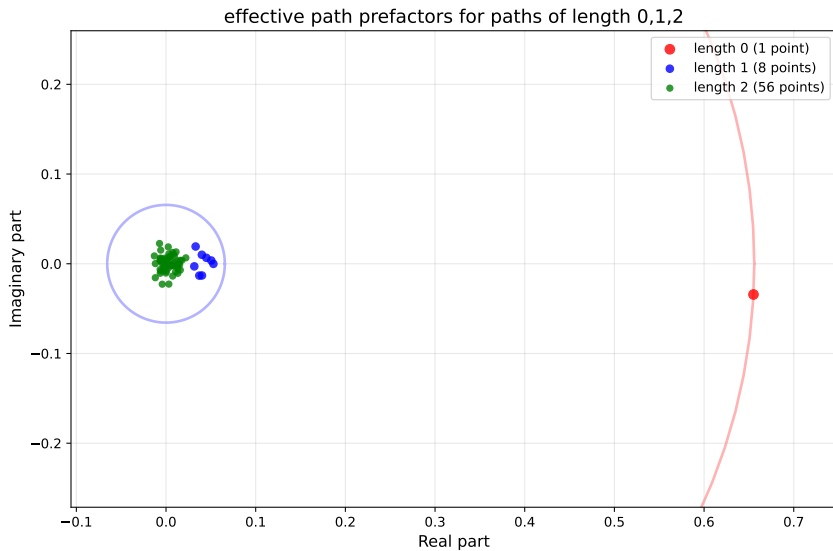
MG

8 layers

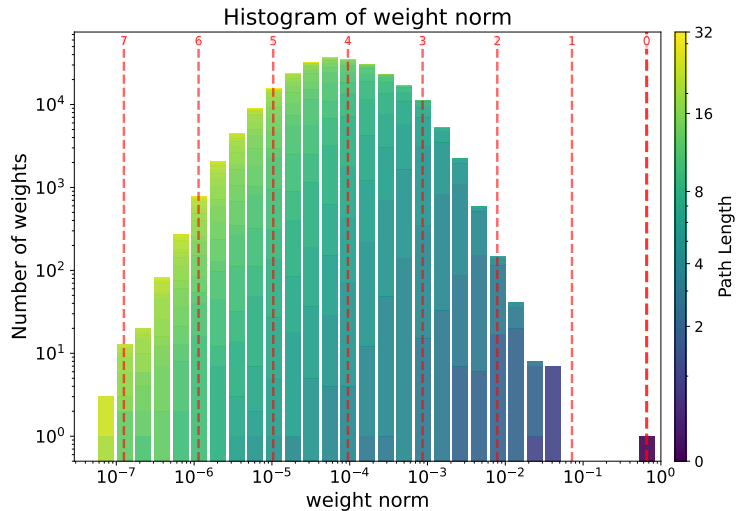
16 layers



# Weights analysis

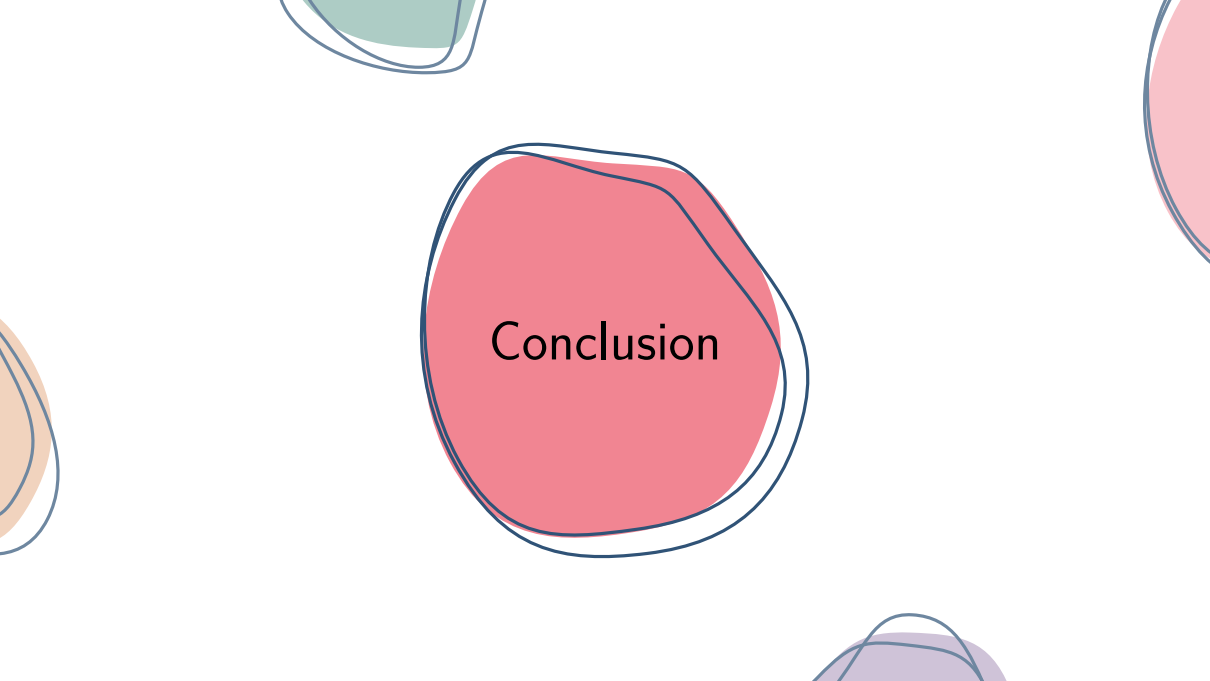


# Weights analysis



Weights look systematic,  
but not like hopping  
expansion!

→ new analytic formula  
for  $D_{WC}^{-1}$ ?



Conclusion



# Conclusion

## Goals

- ☐ Build a preconditioner using ML that mitigates critical slowing down
- ☐ Learn a general analytic formula  
→ No costly setup/trivial transfer
- ☐ Handle topological modes nicely



**Slides and code:**  
**[simon-pfahler.github.io](https://simon-pfahler.github.io)**

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## Goals

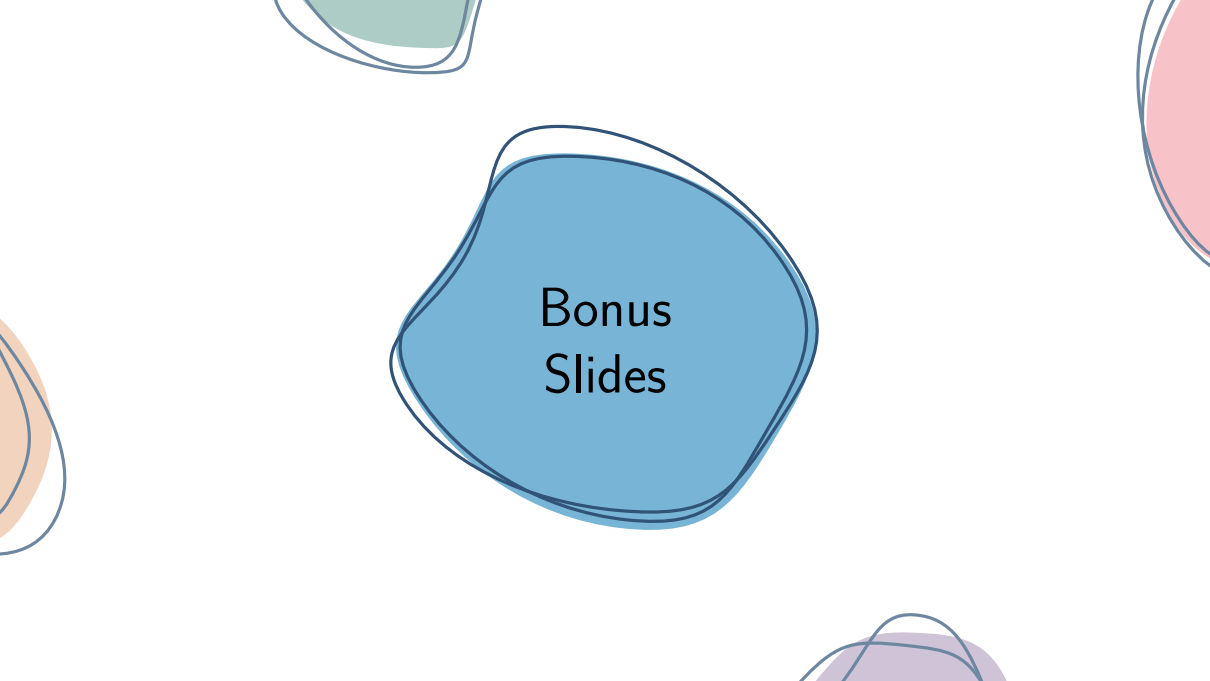
- ✓ Build a preconditioner using ML that mitigates critical slowing down
- ✓ Learn a general analytic formula  
→ No costly setup/trivial transfer
- ✗ Handle topological modes nicely

## Next steps

- ▶ Extract the analytic expansion the models learned
- ▶ Understand why the architecture fails for topological modes
- ▶ Explore models to generate representative near-null space vectors economically



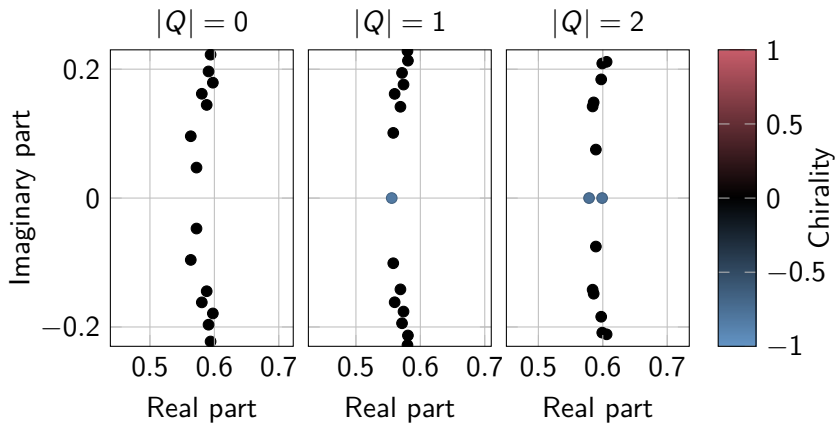
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# Bonus Slides

# Spectra

$8^3 \times 16$  lattice



# Spectra

$16^3 \times 32$  lattice

